

System On Module | SOM

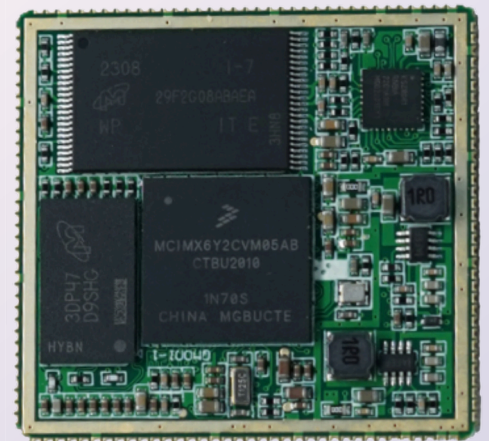
SLS-SoM-MX6



Cost-effective, Power efficient SLS-SoM-MX6 System on Module, providing high computing power and designed for industrial automation & IoT applications development.

The SLS-SoM-i.MX6ULL is a High Performance, Ultra-Miniature, Soldered System on Module, built on the NXP iMX6ULL applications processor family featuring an advanced implementation of a single Arm Cortex-A7 core, operating at speeds of up to 528 MHz.

It integrates power-management, system control, multimedia, memory, basic security modules and other hardware options along with a variety of connectivity interfaces for optimal flexibility and user experience.



Key Features & Benefits

- Cost-efficient, Low-Power, UltraLite NXP – i.MX6ULL, ARM® Cortex-A7® Processor supports up to 528MHz frequency
- Power Management with both hardware and software support for Low-Power Designs
- Built-in Device Security featuring with ARM TrustZone, AES-128 encryption, e-fuse, RNG, and Secure RTC provides basic security
- 512 MB DDR3L and 256 MB Nand Flash Memory support
- On-Board Ethernet-PHY interface
- On-Board Power Supply Circuit.
- **Compact form-factor:** 39 (L) x 37 (W) x 4 (H) mm, without shield / 39 x 37 x 5 mm (With Shield) for Ultimate Reliability and Design
- 140-pin Edge-plated Hole Expansion Interface with 1.0mm pitch supports:
- **Variety of I/Os:** Dual High-speed USB 2.0 (OTG & Host), Dual LAN, Dual CAN, UART, I2C, SPI, GPIO interface etc. for Interconnectivity.
- **Multimedia Interface:** Camera, LCD, Audio, Video for Graphics & Video Hardware Acceleration
- Yocto (V5.15 kirkstone) Linux (V5.15.71) supports
- **Ready to use Software APIs:** Watchdog, Ethernet, Network & Modem Manager, I2C, GPIO, RS232, RS485, CAN, USB etc.
- 10+ Years global technical support

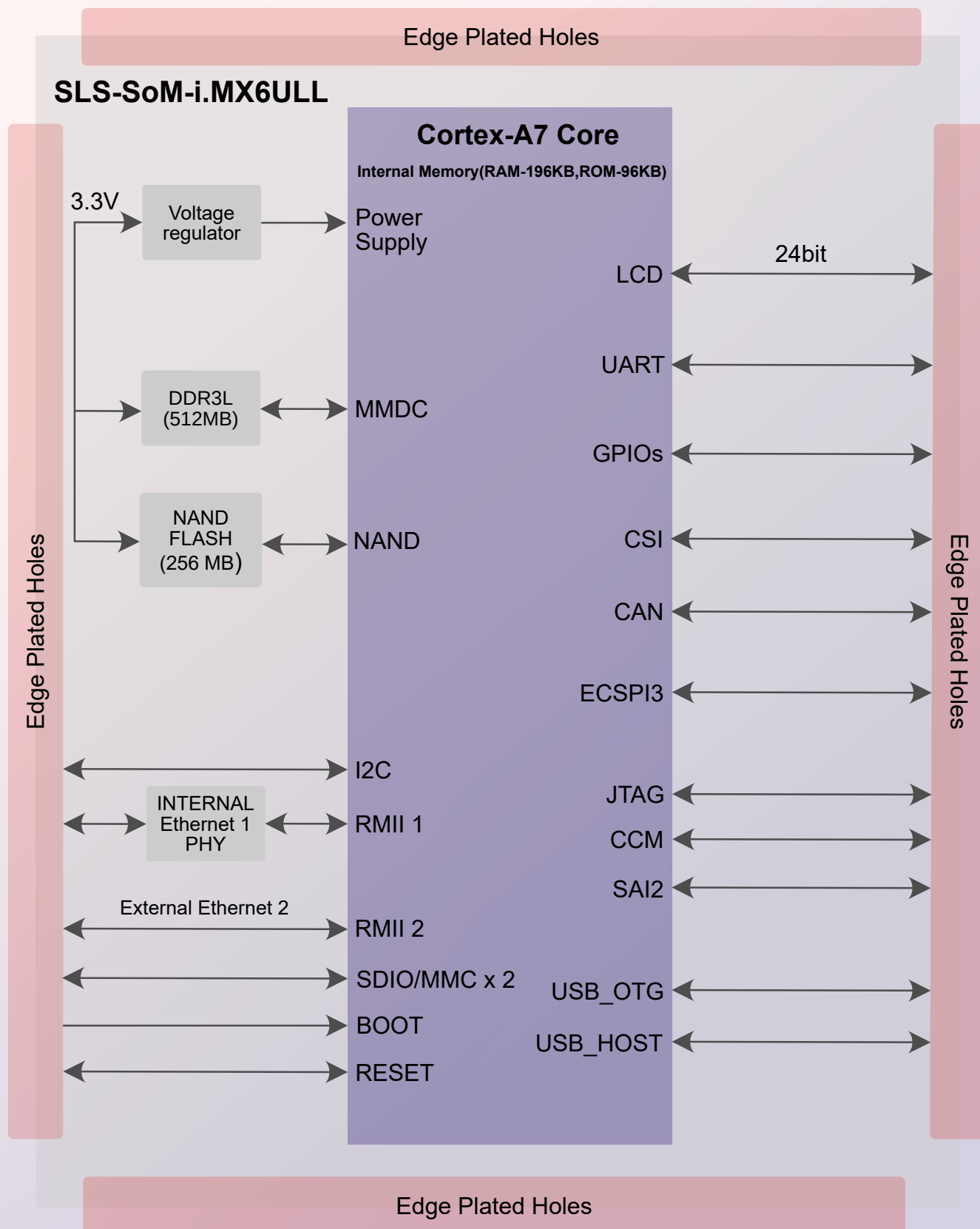
Applications

- Various types of Sensor Interface
- IoT Gateway
- Access Control Panels
- Human Machine Interfaces (HMI)
- Portable Medical and Health Care
- Smart appliances
- eReaders
- Portable Handheld Devices

Specifications

FEATURES	SPECIFICATION	DESCRIPTION
Processor System	CPU Name	NXP i.MX6ULL
	CPU Type	1x Arm Cortex-A7
	CPU Clock	Up to 528MHz
	NEON	Yes
	L1 Instruction Cache	32KB
	L1 Data Cache	32KB
	L2 Cache	128KB
Memory	RAM	512 MB DDR3L, 16 Bit
	Flash	Up to 256 MB NAND Flash
Multimedia	Display Controller	Single
	Digital Audio (I2S/SAI/AC97)	3x up to 1.4 Mbps each
	Parallel RGB	1366 x 768 x 18bpp
	Resistive Touch	4-wire
	Camera Parallel Interface	1x 8/10/16/24 Bit
	LCD (Digital Parallel)	1x 24 Bit
Connectivity	USB 2.0	1x Host , 1x OTG
	Ethernet/Network	2x Ethernet (1x 10/100 MBit/s PHY + 1x RMII)
	I2C	4x @ 400 kbps,
	SPI	4x
	UART	8x
	IrDA	1x
	PWM	8x
	GPIO	up to 97
	Analog Input	10x
	SDIO/SD/MMC	up to 2x
	CAN	2x
	JTAG	1x
Electrical & Environmental	Input Supply Voltage	3.3 VDC, 0.7A
	Power Dissipation	2.3W
	Temperature	-40° to +85° C
Physical	Size	39mm (L) x 37mm (W) x 04 mm (H), without shield
	Expansion Interface	1.0mm pitch, 140 Pin edge plated holes
	Shock / Vibration	Under testing
	PCB Design	8-Layer PCB
	Product Availability	10+ years
	Warrenty	1 Year

Block Daigram



Software (SLS-SoM-YBSP)

Unlike other SOMs, which offers limited deployment-ready-software, SLS-SoM-YBSP integrates a fully tested and validated board support package (BSP) and device drivers with the Yocto Linux OS support for reliable long-term deployment and industrial applications development.

The following are summary of the software features.

Application Software (Customer Specific)
Driver API
Device Drivers
Operating System
Board Support Package (SLS-SoM-YBSP)
Embedded Hardware(SLS-SoM-i.MX6ULL)

Features

ITEM	FEATURE	DESCRIPTION	SOURCE CODE
Bootloader	U-boot	Uboot_2021.10	YES
Linux Kernel	Linux OS	Kernel_5.15.71	YES
Device Driver	USB Host	USB Host driver	YES
	USB OTG	USB OTG driver	YES
	I2C	I2C driver	YES
	SPI	SPI driver	YES
	Ethernet	Ethernet driver	YES
	SDHCI	eMMC/SD Card driver	YES
	Cellular	Cellular	YES
	PWM	PWM Control	YES
	ADC	ADC driver	YES
	RTC	RTC driver	YES
	GPIO	GPIO driver	YES
	UART	UART driver	YES
	CAN	CAN driver	YES
	RS485	Modbus driver	YES
File System	SLS-Image-Core	Customized SLS BSP (Image without GUI interface built with Yocto)	YES
	IMX-Image-Core	IMX default BSP (Image without GUI interface built with Yocto)	YES

Advantages

1. The combination of hardware and software design significantly reduces your complexity of custom board designs risk, development time, and time to market.
2. Due to its cost-optimized bill of material and single-sided assembly, the solderable SLS-SoM-i.MX6ULL module is inexpensive and ideal for industrial use and cost effective high-volume production.
3. The board exhibits readiness for operation across a wide industrial operating temperature range, spanning from -40 to +85 degrees Celsius.
4. Ideal for a wide range of applications development such as Connected devices, IoT Gateway, Access control panels, Human Machine Interfaces (HMI), Portable medical and health care, Smart appliances, eReaders, portable handheld devices and more others

Support	Deliverables	Custom Development (Paid Services)
SLS SOM YBSP	SLS-SoM-MX6ULL System on Module	Custom SOM / Carrier Board Development*
Hardware Design Support (SOM 3D STEP, Library Symbol, CAD Symbol)	SOM User Manual	Hardware Design Review and Support
	Linux User Manual	BSP Development/OS Porting
		Custom Application/GUI Development
		IoT Integration
		Certifications
		*Please contact us for other configurations

About SLS

System Level Solutions (SLS) is ISO9001:2015 and ISO 27001 certified company. SLS is a turnkey product and service provider to the leading private and government enterprises across America, Europe, and Asia. With a chief focus on business-boosting and technology-driven innovative solutions, we act as a technology partners in growth, providing crams solutions across a wide array of industries.

We have in-house capability to rapidly develop, prototype, provide proof of concept and mass production services across hardware, software, firmware, and wireless technologies. Our motive is to work hand-in-hand with your business goals to deliver market-ready products boasting class-apart design, innovative functionalities, excellent quality.

SLS-SoM-MX6ULL System On Module

Scan QR Code
For SLS V-Card



This device can be ordered online from the SLS Website

<https://core.slscorp.com/products/SoM/SLS-SoM-MX6ULL.html>



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